Assignment 4

Econ 8007 Macroeconomic Theory I

October 20, 2026

Question 1

1. Let b_t be the debt-to-output ratio, s_t the primary-surplus-to-output ratio, r the real interest rate, and g the growth rate of output. Starting from the government flow budget constraint in levels, derive
$$b_{t+1} = \frac{1+r}{1+g}\, b_t - s_t.$$
Compute the constant primary surplus that holds b fixed at 1.0 when $(r, g) = (4\%, 2\%)$, and again when $(r, g) = (1.5\%, 2\%)$. State in one sentence what changes economically between the two cases, and why r should not be treated as a constant when b changes.

2. Consider a two-period endowment economy with a continuum of households. All households receive endowments (y_1, y_2) and can save at rate r; a fraction μ additionally faces the borrowing constraint $a_2 \geq 0$, which binds for them (they would like to borrow). The government cuts period-1 lump-sum taxes by Δ for everyone and finances the cut with debt, repaying in period 2 with a lump-sum tax of $(1+r)\Delta$ on everyone.

   (a) Derive each type's consumption response in both periods, and show that aggregate consumption responds by $\Delta C_1 = \mu\Delta$ and $\Delta C_2 = -\mu(1+r)\Delta$.

   (b) The Ricardian equivalence on the slides iterates on the household's flow budget constraint into a lifetime budget constraint and substituting the government's own budget. Identify the step that fails for the constrained households, and explain in one paragraph why the tax cut is, for them, equivalent to a loan from the government.

   (c) Suppose instead the period-2 repayment uses a proportional labor tax (endowments become labor income with an hours choice). YOU DO NOT NEED TO SOLVE THIS CASE. Just name the second channel through which equivalence now fails.

Question 2

A representative household has preferences $\log c + \alpha \log(1-h)$; technology is Cobb–Douglas with capital share θ; labor faces an effective marginal earnings tax τ, with revenue rebated

lump-sum. Derive

$$h = \frac{(1-\tau)(1-\theta)}{(1-\tau)(1-\theta) + \alpha\,(c/y)}.$$

With $\theta = 0.3224$, $\alpha = 1.54$, $c/y = 0.75$, compute h at $\tau = 0.40$ (U.S.) and $\tau = 0.60$ (France) and report the implied hours ratio.

The uncompensated wage elasticity of labor supply with these preferences is zero (because income and substitution effects cancel each other). Explain why the *tax* experiment in the previous part still produces a large hours response. What feature of the experiment makes the relevant elasticity a compensated one? What would happen to the predicted hours gap if, instead of being rebated, the revenue were spent on transfers conditional on not working?

Question 3

This problem illustrates the concept of a transition in an example where a closed-form solution exists. A representative household with $\sum_t \beta^t \ln c_t$; technology $y = k^\theta$ with *full depreciation* ($\delta = 1$); competitive factor markets, so $r_t = \theta k_t^{\theta-1}$ (gross-of-depreciation rental) and $w_t = (1-\theta)k_t^\theta$. The government taxes capital income at rate τ and rebates the revenue lump-sum, so the household budget is $c_t + k_{t+1} = (1-\tau)r_t k_t + w_t + T_t$ with $T_t = \tau r_t k_t$. Parameters: $\beta = 0.96$, $\theta = 0.36$.

1. Guess $k_{t+1} = s\,k_t^\theta$ and verify from the Euler equation that

   $$s = \beta\,\theta\,(1-\tau).$$

   Derive the steady states $k^{ss}(\tau) = s^{1/(1-\theta)}$ and $c^{ss}(\tau) = (1-s)\,(k^{ss})^\theta$. Show further that $\ln k_{t+1} = \ln s + \theta \ln k_t$ is *exactly* log-linear, so along any transition the log-capital gap from the new steady state decays by the factor θ each period. Report k^{ss} and c^{ss} at $\tau = 0.2$ and $\tau = 0$.

2. The economy sits in the $\tau = 0.2$ steady state. At $t = 0$, unanticipated and permanent: $\tau = 0$. Using the closed form, compute the exact paths of k_t, c_t, r_t, w_t for $t = 0, \ldots, 60$. Report the impact response of consumption (sign and percent) and reconcile it in one sentence with the fact that the new steady state has *higher* consumption.

3. Now pretend you do not know the closed form, and compute the same transition with algorithm discussed in the lecture:

   (a) Choose $T = 60$. Guess a path $\{k_t\}_{t=0}^T$ from $k^{ss}(0.2)$ to $k^{ss}(0)$ (e.g. linear); hold k_0 fixed at $k^{ss}(0.2)$.

   (b) *Prices from the guess*: $r_t = \theta k_t^{\theta-1}$.

   (c) *Backward pass*: impose the terminal condition $c_T = c^{ss}(0)$ and iterate the Euler equation backward, $c_t = c_{t+1} / \left[\beta(1-\tau')r_{t+1}\right]$ with $\tau' = 0$, to obtain $\{c_t\}$.

   (d) *Forward pass*: from k_0, iterate the resource constraint $\hat{k}_{t+1} = k_t^\theta - c_t$ to obtain the implied capital path. Guard against negativity: $\hat{k}_{t+1} = \max(\cdot,\ 10^{-6})$.

   (e) *Update with dampening*: $k_t^{new} = \xi \hat{k}_t + (1-\xi)k_t$ with $\xi = 0.1$; iterate to $\max_t |\hat{k}_t - k_t| < 10^{-10}$.

Report the number of iterations, plot the converged path against the exact path from (2), and report the maximum absolute deviation (you should achieve something on the order of 10^{-10}). Then rerun with $\xi = 1$ (no dampening) and without the negativity guard, and describe in one paragraph what happens and why we emphasized both in class.